

Dynamical-Spacetime Discrete-Ordinates Radiation in AthenaK

Circumbinary radiation implementation draft

April 28, 2026

Abstract

We describe a standalone discrete-ordinates radiation module for AthenaK that can evolve radiation on time-dependent ADM backgrounds. The existing radiation solver is tied to stationary Cartesian Kerr-Schild metrics: it stores a signed Killing-energy-normalized intensity, uses metric-dependent angular drift coefficients that can be precomputed, and is therefore efficient for isolated stationary black holes. These assumptions are not appropriate for binary black hole backgrounds or for metric data supplied by a dynamical spacetime solver. The new `dyn_radiation` module keeps a CKS compatibility mode for stationary-solver regression tests, but adds an ADM formulation in which the evolved quantity is the Eulerian densitized intensity, the spatial transport speed is $\alpha s^i - \beta^i$, angular geodesic advection is computed from the ADM Hamiltonian, and the geometric energy source is written in Valencia form using α , β^i , γ_{ij} , and K_{ij} . This avoids assuming a time-independent metric and does not require explicit gauge-field time derivatives beyond the ADM metric-evolution relation for $\partial_t \gamma_{ij}$. The per-angle ADM metric source is applied through the closed-form exponential stage increment for the local multiplicative source. In the present implementation the radiation field is metric-passive: it is transported on ADM/Z4c geometry, but its stress-energy is not included in the Z4c metric right-hand side. We also add a diagnostic null-geodesic particle pusher for beam-edge tracers on ADM backgrounds. Flat-space beam tests reproduce the legacy solver to roundoff, radiation-hydrodynamic linear-wave tests reproduce the legacy error norms in stationary Kerr-Schild mode, and a superposed boosted-Kerr-Schild binary test exercises the ADM beam source, ADM matter coupling, and null-particle tracers.

Keywords: numerical relativity; radiation hydrodynamics; general relativistic radiation transport; binary black holes; AthenaK

1 Introduction

Radiation transport in relativistic accretion flows is usually implemented on a fixed analytic background. For a single Kerr black hole this is a good approximation in many applications: the metric is stationary, the coordinate tetrad can be generated once, and the angular advection terms that bend rays in momentum space can be precomputed. Circumbinary and dynamical-spacetime applications remove these simplifications. The lapse, shift, spatial metric, and extrinsic curvature may change in time; the background may be supplied by a numerical relativity evolution; and the same radiation module should be usable with superposed black-hole data, Z4c data, or any other ADM source.

The goal of the implementation described here is narrower than a full production radiation scheme. We add a separate `dyn_radiation` package, rather than changing the existing `radiation` module, so that the legacy stationary path remains available and bitwise comparisons remain

possible. The current ADM branch implements spatial transport, angular geodesic advection, radiation-matter coupling, and the radiation energy source in a Valencia-compatible form. The angular advection coefficients are rebuilt from cached ADM geometry each stage, so ADM mode now supports `angular_fluxes=true`. Radiation is deliberately one-way coupled to the metric in this pass: the metric can affect radiation through the ADM fields, but radiation does not source the Z4c evolution. The implementation remains separate from the legacy solver so stationary CKS behavior can still be tested directly.

The paper is organized as follows. Section 2 fixes notation and derives the radiation variables from the invariant distribution function. Section 3 summarizes the stationary-metric solver. Section 4 derives the ADM variables, geometric source terms, and angular geodesic operator used by the new solver. Section 5 describes the numerical algorithm and implementation. Section 6 documents the null-particle beam-edge diagnostic. Section 7 presents the regression and smoke tests. Section 9 summarizes direct consistency checks. Section 10 summarizes limitations and next implementation steps.

2 Conventions and Radiation Variables

We use units with $c = G = 1$ and metric signature $(-+++)$. Greek indices $\mu, \nu, \dots$ run over spacetime components 0, 1, 2, 3, Latin indices $i, j, \dots$ run over spatial components 1, 2, 3, and hatted Latin indices $\hat{a}, \hat{b}, \dots$ denote an orthonormal spatial tetrad. Repeated indices are summed. The radiation module is grey: it evolves intensity integrated over photon frequency, but still resolves propagation direction on a discrete angular grid.

The invariant starting point is the photon distribution function $f(x^\mu, p_i)$ on the null mass shell. In vacuum, Liouville's theorem says that f is constant along null geodesics:

$$\frac{df}{d\lambda} = p^\mu \frac{\partial f}{\partial x^\mu} + \frac{dp_i}{d\lambda} \frac{\partial f}{\partial p_i} = 0, \quad (1)$$

where λ is an affine parameter and p_μ is the photon covector. Matter interaction adds an invariant collision term to the right-hand side. For a local orthonormal observer measuring photon frequency ν , the frequency-dependent specific intensity satisfies

$$\frac{I_\nu}{\nu^3} \propto f. \quad (2)$$

After integrating over frequency, the bolometric grey intensity transforms as

$$I \propto \nu^4 f. \quad (3)$$

This is why the matter-coupling kernel converts Eulerian-frame intensity to fluid-comoving intensity with a fourth power of the frequency ratio.

The angular grid stores a finite set of unit directions $\ell_n^{(\hat{a})}$ and quadrature weights w_n , with

$$\ell_n^{(\hat{a})} \ell_{n(\hat{a})} = 1, \quad \sum_n w_n \simeq 4\pi. \quad (4)$$

The continuum angular integrals are approximated by these quadrature sums. The code variable U_n is not the invariant distribution function itself; it is a finite-volume conserved density whose precise normalization depends on the metric formulation. The CKS compatibility path keeps the historical signed stationary-metric normalization. The ADM path uses the Eulerian volume density $U_n = \sqrt{\gamma} I_n$, derived below.

3 Stationary-Metric Radiation Solver

The existing AthenaK radiation package evolves grey specific intensity on a set of angular bins. For each angular bin n , a local orthonormal tetrad direction is

$$q_n^{(\hat{\alpha})} = (1, \ell_n, m_n, p_n), \quad \ell_n^2 + m_n^2 + p_n^2 = 1. \quad (5)$$

The stationary Cartesian Kerr-Schild (CKS) tetrad maps this direction to a coordinate null vector,

$$k_n^\mu = e_{(\hat{\alpha})}^\mu q_n^{(\hat{\alpha})}, \quad k_{\mu,n} = g_{\mu\nu} k_n^\nu. \quad (6)$$

For a stationary metric, the positive photon Killing energy is $-k_\mu(\partial_t)^\mu = -k_0$ for the $(-+++)$ signature used here. The legacy finite-volume variable stores the signed covariant time-component normalization,

$$U_n = k_n^0 k_{0,n} I_n, \quad (7)$$

and the reconstructed primitive entering the spatial flux is

$$k_{0,n} I_n = \frac{U_n}{k_n^0}. \quad (8)$$

The corresponding spatial flux in direction i is

$$F_n^i = k_n^i k_{0,n} I_n. \quad (9)$$

Thus the coordinate transport speed of the legacy conserved variable is $F_n^i/U_n = k_n^i/k_n^0$, while k_0 itself is only the covariant time component; $-k_0$ is the positive Killing energy. Ignoring matter coupling for the moment, the schematic update is

$$\partial_t U_n + \partial_i F_n^i + \nabla_\Omega \cdot A_n = S_n, \quad (10)$$

where A_n is the angular-space flux induced by geodesic bending. In the stationary CKS module the tetrad, angular face maps, and angular drift coefficients are precomputed. This is efficient and accurate on a fixed black hole background, but it encodes the assumptions that the metric is known analytically, stationary, and tied to the CKS tetrad construction.

4 ADM Formulation for Dynamical Backgrounds

4.1 ADM split and Eulerian observers

The dynamical branch uses the standard ADM line element

$$ds^2 = -\alpha^2 dt^2 + \gamma_{ij}(dx^i + \beta^i dt)(dx^j + \beta^j dt), \quad (11)$$

where α is the lapse, β^i is the shift, and γ_{ij} is the spatial metric. The Eulerian normal observer has

$$n^\mu = \left(\frac{1}{\alpha}, -\frac{\beta^i}{\alpha} \right), \quad n_\mu = (-\alpha, 0, 0, 0). \quad (12)$$

The determinant of the spatial metric is denoted γ , so the proper spatial volume element is $\sqrt{\gamma} d^3x$.

The radiation tetrad timelike leg is chosen to be the Eulerian observer,

$$e_{(\hat{0})}^\mu = n^\mu, \quad (13)$$

and define spatial tetrad columns $E_{(\hat{a})}^i$ satisfying

$$e_{(\hat{a})}^\mu = (0, E_{(\hat{a})}^i), \quad \gamma_{ij} E_{(\hat{a})}^i E_{(\hat{b})}^j = \delta_{\hat{a}\hat{b}}. \quad (14)$$

The dual spatial co-triad is $e_i^{(\hat{a})}$, with

$$e_i^{(\hat{a})} E_{(\hat{b})}^i = \delta_{\hat{b}}^{\hat{a}}, \quad \gamma_{ij} = e_i^{(\hat{a})} e_{(\hat{a})j}. \quad (15)$$

For angular-bin direction $\ell_n^{(\hat{a})}$, the Eulerian spatial unit vector in coordinates is

$$s_n^i = E_{(\hat{a})}^i \ell_n^{(\hat{a})}, \quad \gamma_{ij} s_n^i s_n^j = 1. \quad (16)$$

4.2 Densitized intensity

For a photon with Eulerian frequency

$$\nu_E = -n_\mu p^\mu = \alpha p^0, \quad (17)$$

write the photon four-momentum as

$$p^\mu = \nu_E (n^\mu + s^\mu), \quad s^\mu = (0, s^i), \quad (18)$$

where $s^\mu n_\mu = 0$ and $\gamma_{ij} s^i s^j = 1$. Equation (18) immediately gives the coordinate propagation speed

$$v_n^i = \frac{p^i}{p^0} = \alpha s_n^i - \beta^i. \quad (19)$$

The ADM conserved variable is the Eulerian proper-volume density of grey intensity,

$$U_n = \sqrt{\gamma} I_n, \quad (20)$$

with spatial flux

$$F_n^i = \sqrt{\gamma} v_n^i I_n. \quad (21)$$

Equations (19)–(21) reduce to ordinary flat-space transport when $\alpha = 1$, $\beta^i = 0$, and $\gamma_{ij} = \delta_{ij}$. They also make the normalization difference from the legacy solver explicit: ADM mode evolves the volume-weighted intensity seen by Eulerian observers, not the Killing-energy weighted intensity in Equation (7).

4.3 Radiation moments and Valencia equations

The stress-energy tensor of a grey radiation field can be decomposed relative to the Eulerian observer as

$$T_{\text{rad}}^{\mu\nu} = E n^\mu n^\nu + n^\mu F^\nu + F^\mu n^\nu + P^{\mu\nu}, \quad (22)$$

where $F^\mu n_\mu = 0$ and $P^{\mu\nu} n_\nu = 0$. In the tetrad quadrature used by the code,

$$E = \sum_n w_n I_n, \quad (23)$$

$$F^i = \sum_n w_n I_n s_n^i, \quad (24)$$

$$P^{ij} = \sum_n w_n I_n s_n^i s_n^j, \quad (25)$$

and $P^i_j = P^{ik}\gamma_{kj}$. These definitions are useful for verification: summing the discrete per-angle equations must recover the conservative equations for E and $S_i = \gamma_{ij}F^j$.

Starting from stress-energy conservation,

$$\nabla_\mu T^{\mu\nu}_{\text{rad}} = G^\nu, \quad (26)$$

where G^ν is the force density on the radiation field. The force on the matter is therefore $-G^\nu$. With this sign convention, the Eulerian energy equation is obtained by projecting along n_ν . In vacuum ($G^\nu = 0$), or before applying matter coupling, this gives

$$\partial_t(\sqrt{\gamma}E) + \partial_i[\sqrt{\gamma}(\alpha F^i - \beta^i E)] = \sqrt{\gamma}(\alpha P^{ij}K_{ij} - F^i\partial_i\alpha). \quad (27)$$

Projecting Equation (26) into the spatial slice gives the momentum equation

$$\partial_t(\sqrt{\gamma}S_j) + \partial_i[\sqrt{\gamma}(\alpha P^i_j - \beta^i S_j)] = \sqrt{\gamma}\left[\frac{1}{2}\alpha P^{ik}\partial_j\gamma_{ik} + S_i\partial_j\beta^i - E\partial_j\alpha\right]. \quad (28)$$

Equations (27) and (28) are the same geometric source structure used by Valencia GR hydrodynamics. Metric time evolution enters through the ADM identity for $\partial_t\gamma_{ij}$, with K_{ij} supplying the energy-source term and the co-triad time derivative supplying the angular drift. This is the central reason the ADM radiation path is not tied to a stationary gauge.

For a single angular bin, substitute

$$E \rightarrow I_n, \quad F^i \rightarrow I_n s_n^i, \quad P^{ij} \rightarrow I_n s_n^i s_n^j.$$

The per-angle geometric energy source used in the code is therefore

$$S_n^{\text{geom}} = U_n(\alpha K_{ij}s_n^i s_n^j - s_n^i\partial_i\alpha). \quad (29)$$

Summing Equation (29) over angles reproduces the right-hand side of Equation (27). The energy source is agnostic to the gauge condition used to obtain the ADM data and does not require assuming $\partial_t\alpha = 0$, $\partial_t\beta^i = 0$, or $\partial_t\gamma_{ij} = 0$.

4.4 ADM angular geodesic advection

The angular advection operator accounts for geodesic bending, i.e. the change of $\ell^{(\hat{a})}$ as a ray moves through a curved or time-dependent tetrad. It is derived from the null Hamiltonian. The inverse ADM metric is

$$g^{00} = -\frac{1}{\alpha^2}, \quad g^{0i} = \frac{\beta^i}{\alpha^2}, \quad g^{ij} = \gamma^{ij} - \frac{\beta^i\beta^j}{\alpha^2}. \quad (30)$$

The null constraint $g^{\mu\nu}p_\mu p_\nu = 0$ gives

$$-p_0 = H(x^i, p_i) = \alpha\Pi - \beta^i p_i, \quad \Pi = \sqrt{\gamma^{ij}p_i p_j}. \quad (31)$$

Using coordinate time t as the evolution parameter, Hamilton's equations are

$$\dot{x}^i = \frac{\partial H}{\partial p_i} = \alpha \frac{\gamma^{ij}p_j}{\Pi} - \beta^i, \quad (32)$$

$$\dot{p}_i = -\frac{\partial H}{\partial x^i} = -(\partial_i\alpha)\Pi + p_j\partial_i\beta^j - \frac{\alpha}{2\Pi}p_j p_k \partial_i\gamma^{jk}. \quad (33)$$

The code normalizes the momentum by the Eulerian photon energy, so $\Pi = 1$,

$$p_i = e^{(\hat{a})}_i \ell_{(\hat{a})}, \quad s^i = \gamma^{ij} p_j = E^i_{(\hat{a})} \ell^{(\hat{a})}. \quad (34)$$

With this normalization, the coordinate speed follows from Equations (31) and (33). The force used to advance the angular direction is the energy-rescaled canonical force $A_i = (1/\nu_E) dP_i/dt$ for the unnormalized covariant momentum $P_i = \nu_E p_i$:

$$H(x^i, p_i) = \alpha \sqrt{\gamma^{ij} p_i p_j} - \beta^i p_i. \quad (35)$$

For the Eulerian tetrad normalization used here, $\sqrt{\gamma^{ij} p_i p_j} = 1$, $p_i = e^{(\hat{a})}_i \ell_{(\hat{a})}$, and

$$\dot{x}^i = \alpha s^i - \beta^i, \quad (36)$$

$$A_i = -\partial_i \alpha + p_j \partial_i \beta^j - \frac{\alpha}{2} p_j p_k \partial_i \gamma^{jk}. \quad (37)$$

The literal derivative of the normalized covector $p_i = P_i/\nu_E$ differs from A_i by a component parallel to p_i , which carries the finite gravitational/Doppler frequency shift. The angular update below removes that parallel component and keeps only the tangential drift on the unit sphere.

The angular direction is obtained from the co-momentum through

$$\ell^{(\hat{a})} = E^i_{(\hat{a})} p_i. \quad (38)$$

Differentiating in the same triad gauge gives

$$\dot{\ell}^{(\hat{a})} = E^i_{(\hat{a})} \left[A_i - \left(\partial_t e^{(\hat{b})}_i + \dot{x}^j \partial_j e^{(\hat{b})}_i \right) \ell_{(\hat{b})} \right]. \quad (39)$$

Any component of $\dot{\ell}^{(\hat{a})}$ parallel to $\ell^{(\hat{a})}$ is the frequency/normalization drift of the unconstrained canonical force, not a physical angular drift on the unit sphere, and is projected away:

$$\dot{\ell}^{(\hat{a})} \leftarrow \dot{\ell}^{(\hat{a})} - (\ell_{\hat{b}} \dot{\ell}^{(\hat{b})}) \ell^{(\hat{a})}. \quad (40)$$

The time derivative of the co-triad is obtained by differentiating the same Cholesky factorization used for the spatial triad, with

$$\partial_t \gamma_{ij} = -2\alpha K_{ij} + D_i \beta_j + D_j \beta_i. \quad (41)$$

This relation is an identity in the ADM split and is independent of the particular gauge condition used to evolve α and β^i . The implementation uses the equivalent partial-derivative form

$$\partial_t \gamma_{ij} = -2\alpha K_{ij} + \beta^k \partial_k \gamma_{ij} + \gamma_{ik} \partial_j \beta^k + \gamma_{jk} \partial_i \beta^k, \quad (42)$$

which makes the shift Lie derivative explicit.

4.5 Cholesky tetrad gauge and derivatives

The implementation uses a Cholesky gauge for the spatial tetrad. Let $\gamma_{ij} = L_{i\hat{a}} L_{j\hat{a}}$, with L lower triangular:

$$L_{1\hat{1}} = \sqrt{\gamma_{11}}, \quad L_{2\hat{1}} = \frac{\gamma_{12}}{L_{1\hat{1}}}, \quad L_{3\hat{1}} = \frac{\gamma_{13}}{L_{1\hat{1}}}, \quad (43)$$

$$L_{2\hat{2}} = \sqrt{\gamma_{22} - L_{2\hat{1}}^2}, \quad L_{3\hat{2}} = \frac{\gamma_{23} - L_{3\hat{1}} L_{2\hat{1}}}{L_{2\hat{2}}}, \quad L_{3\hat{3}} = \sqrt{\gamma_{33} - L_{3\hat{1}}^2 - L_{3\hat{2}}^2}. \quad (44)$$

The co-triad is $e_i^{(\hat{a})} = L_{i\hat{a}}$, and the triad columns are $E^i_{(\hat{a})} = (L^{-T})^i_{\hat{a}}$. For any derivative operator ∂_q , where q may be a spatial coordinate or t , the code uses the differentiated Cholesky relations

$$\partial_q L_{1\hat{1}} = \frac{\partial_q \gamma_{11}}{2L_{1\hat{1}}}, \quad (45)$$

$$\partial_q L_{2\hat{1}} = \frac{\partial_q \gamma_{12} - L_{2\hat{1}} \partial_q L_{1\hat{1}}}{L_{1\hat{1}}}, \quad (46)$$

$$\partial_q L_{3\hat{1}} = \frac{\partial_q \gamma_{13} - L_{3\hat{1}} \partial_q L_{1\hat{1}}}{L_{1\hat{1}}}, \quad (47)$$

$$\partial_q L_{2\hat{2}} = \frac{\partial_q \gamma_{22} - 2L_{2\hat{1}} \partial_q L_{2\hat{1}}}{2L_{2\hat{2}}}, \quad (48)$$

$$\partial_q L_{3\hat{2}} = \frac{\partial_q \gamma_{23} - (\partial_q L_{3\hat{1}}) L_{2\hat{1}} - L_{3\hat{1}} (\partial_q L_{2\hat{1}}) - L_{3\hat{2}} (\partial_q L_{2\hat{2}})}{L_{2\hat{2}}}, \quad (49)$$

$$\partial_q L_{3\hat{3}} = \frac{\partial_q \gamma_{33} - 2L_{3\hat{1}} \partial_q L_{3\hat{1}} - 2L_{3\hat{2}} \partial_q L_{3\hat{2}}}{2L_{3\hat{3}}}. \quad (50)$$

These equations are important for reproducibility: the angular drift uses the same tetrad gauge as the spatial transport and matter-coupling transforms.

4.6 Projection to the angular finite-volume grid

The geodesic angular mesh is a finite-volume mesh on the unit sphere. Writing

$$\ell = (\sin \theta \cos \psi, \sin \theta \sin \psi, \cos \theta),$$

the tangent angular velocities are

$$\dot{\theta} = -\frac{\dot{\ell}_z}{\sin \theta}, \quad (51)$$

$$\sin^2 \theta \dot{\psi} = \ell_x \dot{\ell}_y - \ell_y \dot{\ell}_x. \quad (52)$$

The code projects these two components onto the precomputed angular face normal stored by the geodesic grid. The resulting scalar face speed is the array `na`. In ADM mode `na` is already a coordinate-time angular velocity, so the angular finite-volume flux upwinds U_n directly. In the CKS compatibility branch the historical angular primitive U_n/k_n^0 is still used, preserving the stationary solver behavior. The spherical-coordinate expressions are singular at the poles; the implementation treats them only as a local representation before projection to geodesic-grid face normals and suppresses the polar singular limit rather than using a divergent $\dot{\theta}$.

5 Numerical Algorithm and Implementation

5.1 Standalone solver layout

The new module is intentionally independent of `src/radiation`. It lives under `src/dyn_radiation` and supports two metric modes:

CKS compatibility mode uses the same normalization and transport variables as the legacy solver, enabling direct regression comparisons.

ADM mode uses Equations (20)–(29) and obtains α , β^i , γ_{ij} , and K_{ij} from the active relativistic mesh data.

The solver is constructed as a relativistic component when either the CKS path or the dynamical ADM path is active. This is required so that the radiation state has access to the metric-dependent arrays used by the flux and source kernels.

For Z4c evolutions the present solver is a test-field radiation system. The NR task graph no longer queues a radiation stress-energy task before `Z4c.CalcRHS`, and `Tmunu` storage is not allocated solely because `dyn_radiation` is enabled. Matter stress-energy from the MHD/dynGRMHD path is unchanged. A run that combines `dyn_radiation` and `z4c` must use `geometry=adm`; the CKS mode is retained only for fixed background compatibility and regression tests.

5.2 Finite-volume equation actually advanced

For each angular cell n , ADM mode advances

$$\partial_t U_n + \partial_i (\sqrt{\gamma} v_n^i I_n) + \mathcal{D}_\Omega (U_n \ell_n) = S_n^{\text{geom}} + C_n, \quad (53)$$

where $U_n = \sqrt{\gamma} I_n$, $v_n^i = \alpha s_n^i - \beta^i$, $\mathcal{D}_\Omega$ is the finite-volume angular divergence on the geodesic grid, S_n^{geom} is Equation (29), and C_n contains emission, absorption, scattering, and user source terms. On a Cartesian mesh cell with coordinate widths Δx^i , the explicit transport part of one Runge–Kutta substep is

$$U_n^* = U_n^m - \Delta t \sum_i \frac{F_{n,+}^i - F_{n,-}^i}{\Delta x^i} - \Delta t \frac{1}{\Delta \Omega_n} \sum_{f \in \partial \Omega_n} \Delta l_f A_{n,f}, \quad (54)$$

where $F_{n,\pm}^i$ are reconstructed spatial face fluxes, $\Delta \Omega_n = w_n$, Δl_f is the angular edge length, and

$$A_{n,f} = n_{n,f}^a \begin{cases} U_{\text{upwind}}, & \text{ADM mode,} \\ (U/k^0)_{\text{upwind}}, & \text{CKS compatibility mode.} \end{cases} \quad (55)$$

Here $n_{n,f}^a$ is the scalar angular face speed stored in `na`. The distinction in Equation (55) is the main normalization difference between the two metric modes.

The local ADM metric-energy source is multiplicative in the stage-start intensity. Writing

$$G_n = \alpha K_{ij} s_n^i s_n^j - s_n^i \partial_i \alpha, \quad (56)$$

the code applies the source contribution over the RK substep Δt_s as

$$U_n^{**} = U_n^* + U_n^m [\exp(\Delta t_s G_n) - 1], \quad (57)$$

where U_n^m is the current stage value. For the isolated geometric source this is the exact update of $\partial_t U_n = G_n U_n$. In the full method it is an exponential source increment inside an explicit flux-source stage; the combined spatial/angular transport plus source update remains a split finite-volume approximation and is validated by the transport tests.

5.3 Stage update

At each explicit stage, ADM mode performs the following operations:

1. Refresh ADM variables on cell centers and faces from the stage-start geometry.

2. Construct Eulerian spatial tetrads by factorizing γ_{ij} .
3. Fill cached $\sqrt{\gamma}$, α , β^i , γ_{ij} , γ^{ij} , K_{ij} , tetrad columns, co-triads, spatial metric gradients, and co-triad derivatives.
4. Reconstruct $I_n = U_n/\sqrt{\gamma}$ to faces.
5. Compute face speeds $v_n^i = \alpha s_n^i - \beta^i$ and fluxes from Equation (21).
6. Compute angular drift from Equations (37) and (39); form angular fluxes using U_n as the upwinded quantity.
7. Apply flux divergence to U_n .
8. Apply the per-angle geometric source rate in Equation (56) with the exponential increment in Equation (57).
9. Apply matter/source terms, boundary exchange, and AMR operations.

In the NR task graph this refresh is a dedicated `Rad_PrepGeom` task. `Rad_Flux` and the radiation coupling path depend on this cache refresh and therefore use the same stage-start ADM state. `Rad_Flux` does not wait for the post-update `Z4c_Z4c2ADM` conversion. The radiation timestep task may wait for `Z4c_Z4c2ADM`, so the next timestep is limited by the post-stage ADM geometry. This ordering keeps radiation transport consistent with the RK stage while still measuring the next CFL limit on the latest metric state.

The cache is deliberately cell-local and device-resident. The primary state arrays store the ADM fields and co-triad:

```
adm_alpha_c,  adm_beta_u_c,  adm_g_dd_c,
adm_g_uu_c,  adm_K_dd_c,  adm_cotriad_c.
```

The derivative arrays

```
adm_grad_alpha_c,  adm_grad_beta_u_c,
adm_grad_g_dd_c,  adm_grad_g_uu_c,  adm_grad_cotriad_c
```

store centered spatial derivatives, using one-sided differences only at the edge of the allocated block. The array `adm_dt_cotriad_c` stores $\partial_t e_i^{(\hat{a})}$ computed from Equations (41) and (50). These arrays are Kokkos device views; no host containers or dynamic allocation enter the transport kernels.

The ADM timestep limiter has three parts: coordinate spatial speeds, angular-geodesic speeds, and the local geometric source rate G_n . The source-rate cap is no longer needed for linear-source positivity, because the metric source is integrated exponentially, but it still prevents large single-stage amplification factors in strong-field cells. In runs with radiation excision, all three reductions skip cells marked by `rad_mask`; otherwise masked cells with singular or intentionally unphysical metric data can set a meaningless global timestep.

5.4 Beam source and output normalization

The beam source now uses the active radiation tetrad, rather than assuming the legacy CKS metric, to construct the beam angle and source normalization. The increment to the conserved radiation variable is

$$\Delta U_n = \begin{cases} k_n^0 k_{0,n} \dot{I} \Delta t, & \text{legacy or CKS-compatible normalization,} \\ \sqrt{\gamma} \dot{I} \Delta t, & \text{ADM normalization.} \end{cases} \quad (58)$$

The first branch uses the same signed legacy normalization as Equation (7); a positive physical intensity is recovered by the inverse normalization below even though $k^0 k_0 < 0$ for a future-directed flat-space photon in the $(-+++)$ convention. Moment output applies the inverse normalization,

$$I_n = \begin{cases} U_n/(k_n^0 k_{0,n}), & \text{legacy or CKS-compatible normalization,} \\ U_n/\sqrt{\gamma}, & \text{ADM normalization.} \end{cases} \quad (59)$$

This separation is what allows the flat ADM beam test to reproduce the legacy flat CKS beam component by component, even though the conserved variables differ internally.

5.5 Radiation-matter coupling in ADM variables

The matter-coupling routine is local to a cell and follows the legacy grey implicit update, but with ADM frame transformations and moment feedback. The fluid primitive velocity variables are the Valencia spatial four-velocity components $u_{(\text{spatial})}^i \equiv W v_{\text{Eulerian}}^i$. In a shifted coordinate system this equals the Eulerian spatial projection $u^i + W\beta^i/\alpha$, not the bare coordinate component $u^i = dx^i/d\tau$. The Lorentz factor measured by the Eulerian observer is

$$W = \sqrt{1 + \gamma_{ij} u_{(\text{spatial})}^i u_{(\text{spatial})}^j}. \quad (60)$$

The ADM normal-to-tetrad transform maps the fluid four-velocity into the Eulerian orthonormal frame:

$$u^{(\hat{0})} = W, \quad u^{(\hat{a})} = e^{(\hat{a})}_i u_{(\text{spatial})}^i. \quad (61)$$

For a photon direction $(1, \ell^{(\hat{a})})$ in the Eulerian tetrad, the frequency ratio between the fluid-comoving frame and the Eulerian frame is

$$D \equiv \frac{\nu_{\text{cm}}}{\nu_{\text{E}}} = u^{(\hat{0})} - u^{(\hat{a})} \ell_{(\hat{a})}. \quad (62)$$

This is the special-relativistic Doppler factor written in the local Eulerian tetrad. The grey intensity and solid angle used by the coupling update are

$$I_{\text{cm}} = I_{\text{E}} D^4, \quad d\Omega_{\text{cm}} = D^{-2} d\Omega_{\text{E}}, \quad (63)$$

which follows from I_ν/ν^3 invariance and aberration. In ADM mode $I_{\text{E}} = U/\sqrt{\gamma}$. In CKS compatibility mode the legacy normalization $I_{\text{E}} = U/(k^0 k_0)$ is retained. For comoving opacity coefficients per unit comoving length/time, a coordinate substep Δt corresponds to an angle-dependent comoving path length

$$\Delta\tau_n = \alpha D_n \Delta t. \quad (64)$$

This factor is included in the local implicit solve through the ADM lapse and Doppler factor D_n ; flat comoving tests have $\alpha = D_n = 1$ and therefore do not exercise this distinction.

The grey absorption/scattering update can be written schematically as

$$\frac{\Delta I_{\text{cm}}}{\Delta\tau} = \sigma_a(B - I_{\text{cm}}) + \sigma_s(J_{\text{cm}} - I_{\text{cm}}) + \sigma_p(B - J_{\text{cm}}), \quad (65)$$

where $B = a_{\text{rad}} T_{\text{gas}}^4/(4\pi)$, J_{cm} is the comoving-frame angular mean intensity, and $\sigma_a, \sigma_s, \sigma_p$ are the absorption, scattering, and Planck-correction coefficients used by the existing opacity helper. The gas internal energy is linear in T_{gas} for the ideal-gas EOS, while $B \propto T_{\text{gas}}^4$. The local implicit solve therefore reduces to the same quartic temperature equation used by the legacy solver. The new

ADM-specific parts are the frame normalization in Equations (62)–(63) and the feedback moments below. The implementation now iterates this local nonlinear solve up to `source_max_iter` times, with convergence controlled by `source_tolerance`. This is important when the opacity depends on the gas temperature or when the optical depth per step is large. Each sub-iteration rebuilds the opacity-weighted angular coefficients and solves the monotone quartic with a bracketed Newton-bisection method, avoiding the weak-coupling failure mode of the closed-form quartic expression used in the original legacy path.

After explicit transport, angular advection, and local matter coupling, the solver enforces positivity with a zeroth-moment-conserving angular redistribution rather than independent clipping. In ADM mode the limiter acts on the conserved $U_n = \sqrt{\gamma} I_n$. In CKS compatibility mode it acts on the primitive legacy intensity $I_n = U_n / (k^0 k_0)$, since $k^0 k_0$ carries the stationary tetrad sign convention. Negative angular bins are set to zero and the positive bins in the same cell are rescaled so the angular zeroth moment is unchanged whenever the pre-limited angular integral is nonnegative. The limiter is not a momentum-conserving angular remap, and if the whole cell has negative angular energy no nonnegative conservative redistribution exists and the cell is zeroed. Matter-coupling feedback uses the post-limiter radiation moments in that local source step, but limiter activations during transport should still be diagnosed as under-resolution or an overly aggressive reconstruction.

Before and after the implicit radiation update, ADM mode computes the Valencia radiation conserved moments

$$\sqrt{\gamma} E = \sum_n w_n U_n, \quad (66)$$

$$\sqrt{\gamma} S_i = \sum_n w_n U_n \gamma_{ij} s_n^j. \quad (67)$$

The gas conserved variables receive the opposite change,

$$\Delta(\sqrt{\gamma} \tau_{\text{gas}}) = -\Delta(\sqrt{\gamma} E_{\text{rad}}), \quad \Delta(\sqrt{\gamma} S_i^{\text{gas}}) = -\Delta(\sqrt{\gamma} S_i^{\text{rad}}), \quad (68)$$

up to the existing choice of whether `affect_fluid` is enabled. This is the practical consistency condition for local matter coupling: the same Eulerian radiation variables used in the transport equation are used to feed energy and momentum back to the fluid. If `fixed_fluid=true`, the solver treats the gas state as a prescribed background for opacity and comoving-frame transforms. In that mode `affect_fluid` defaults to false, and the constructor rejects the contradictory request `fixed_fluid=true, affect_fluid=true`; otherwise conserved fluid variables could be changed while primitives remain fixed.

5.6 Cost of tetrad construction

The stationary `radiation` module and the ADM branch of `dyn_radiation` make different algorithmic choices because they target different metric regimes. In the stationary CKS solver the metric is analytic and time independent. The setup routine can therefore call the CKS metric, metric-derivative, and tetrad builders at cell centers, again at each coordinate face, and again where angular drift and normal-to-tetrad transforms are needed. Those calls are acceptable because they are made only when the mesh geometry is initialized or rebuilt. The transport kernels then read compact precomputed arrays: the full center tetrad $e_{(\hat{\alpha})}^{\mu}$, its covariant form, and one tetrad column on each face. This is an efficient design for a fixed black-hole background, but it relies on the existence of a stationary analytic metric and on the legacy Killing-energy normalization.

The ADM path cannot use that strategy directly. The metric fields may come from Z4c or from another ADM provider and may change every RK stage. Repeating a general CKS-style tetrad

construction, metric derivative evaluation, and connection construction inside each angle loop would turn geometry preparation into an $O(N_{\text{cell}}N_{\Omega})$ operation and would duplicate work across transport, angular advection, and matter coupling. Instead, `dyn_radiation` pays a structured per-stage cache cost. At the beginning of each stage, `Rad_PrepGeom` refreshes the ADM fields, constructs the Eulerian tetrad, and fills all geometry arrays used by the later kernels. The hot kernels then perform only local array reads and small fixed-size contractions.

The tetrad itself is specialized to the Eulerian ADM split. The timelike leg is the normal observer, $e_{(0)}^{\mu} = (1/\alpha, -\beta^i/\alpha)$, and the spatial legs are obtained from a Cholesky factorization of γ_{ij} . Thus the cell-center construction requires a three-dimensional spatial factorization rather than a general four-dimensional tetrad solve. At faces the code does not store a full face tetrad. It stores only the four transport coefficients needed to compute the coordinate face speed,

$$v_n^d = -\beta^d + \alpha E_{(\hat{a})}^d \ell_n^{(\hat{a})},$$

namely $-\beta^d$ and the three entries $\alpha E_{(\hat{a})}^d$. Each angular bin then forms

$$s_n^i = E_{(\hat{a})}^i \ell_n^{(\hat{a})} \quad (69)$$

or v_n^d with a small dense matrix-vector product.

The same cache also removes repeated geometric work from angular transport and matter coupling. It contains $\sqrt{\gamma}$, α , β^i , γ_{ij} , γ^{ij} , K_{ij} , the triad and co-triad, centered spatial gradients of the ADM fields, and $\partial_t e_i^{(\hat{a})}$. The normal-to-tetrad map used by the radiation-fluid source is especially simple in this gauge: the Eulerian time component maps to itself and the spatial block is just the cached co-triad. Angular geodesic advection reuses the cached gradients and co-triad derivatives to evaluate Equations (37) and (39); it does not reconstruct a four-dimensional connection for every angle.

This is a memory-for-work tradeoff, not a statement that ADM mode is always cheaper. The ADM branch allocates more cell-centered and face-centered geometry arrays than the legacy solver and must refresh them once per stage whenever the metric is dynamical. The benefit is that expensive metric and tetrad operations scale as $O(N_{\text{cell}})$ per stage, while the unavoidable angular work scales as $O(N_{\text{cell}}N_{\Omega})$ with only cached contractions. In CKS compatibility mode `dyn_radiation` deliberately follows the legacy normalization and tetrad construction, so it should be viewed as a regression path rather than an optimized ADM geometry path.

5.7 Code-to-equation map

Table 1 maps the main ADM formulas to the implementation. The purpose is to make review of signs and normalizations straightforward.

6 Null Geodesic Beam-Edge Tracers

The particle module now has an opt-in `null_geodesic` pusher for diagnostic beam-edge trajectories on ADM backgrounds. Particle real data are interpreted as

$$(x, y, z, v_x, v_y, v_z) \longrightarrow (x^i, k_i), \quad (70)$$

where k_i is the covariant spatial photon momentum. The coordinate-time Hamiltonian is

$$\mathcal{H}(x^i, k_i) = \alpha \sqrt{\gamma^{ij} k_i k_j} - \beta^i k_i. \quad (71)$$

Mathematical object	Code storage or routine	Main use
α	<code>adm_alpha.c</code>	speed, source, Hamiltonian
β^i	<code>adm_beta_u.c</code>	$v^i = \alpha s^i - \beta^i$, Hamiltonian
γ_{ij}, γ^{ij}	<code>adm_g_dd.c, adm_g_uu.c</code>	tetrad, momentum, metric gradients
K_{ij}	<code>adm_K_dd.c</code>	Valencia energy source
$E^i_{(\hat{a})}$	<code>tet.c</code> spatial block	coordinate direction s^i
$e^{(\hat{a})}_i$	<code>adm_cotriad.c</code>	momentum p_i , fluid tetrad velocity
$\partial_i \alpha, \partial_i \beta^j, \partial_i \gamma^{jk}$	<code>adm_grad.*</code> arrays	Hamiltonian $\dot{p}_i$, energy source
$\partial_t e^{(\hat{a})}_i$	<code>adm_dt_cotriad.c</code>	tetrad time-dependence in $\dot{\ell}^{(\hat{a})}$
$\dot{\ell} \cdot \hat{n}_\Omega$	<code>na</code>	angular finite-volume flux

Table 1: Code-to-equation map for the ADM radiation implementation.

Let

$$K = \sqrt{\gamma^{ij} k_i k_j}. \quad (72)$$

The pusher advances

$$\frac{dx^i}{dt} = \frac{\partial \mathcal{H}}{\partial k_i} = \alpha \frac{\gamma^{ij} k_j}{K} - \beta^i, \quad (73)$$

$$\frac{dk_i}{dt} = -\frac{\partial \mathcal{H}}{\partial x^i} = -(\partial_i \alpha) K - \frac{\alpha}{2K} (\partial_i \gamma^{jk}) k_j k_k + (\partial_i \beta^j) k_j. \quad (74)$$

The current implementation uses the nearest cell's ADM fields and centered finite differences for the metric derivatives. That accuracy is sufficient for beam-edge diagnostics and visualization. Higher-order interpolation can be added later without changing the particle data layout.

7 Verification and Regression Tests

7.1 Stationary and flat-space beam tests

The first regression compares three beam calculations:

1. the legacy `radiation` solver on a flat CKS background,
2. `dyn_radiation` in CKS compatibility mode, and
3. `dyn_radiation` in ADM flat-space mode.

The reproducibility inputs are stored in

```

dyngr_radiation_paper/inputs/beam_radiation_cks.athinput
dyngr_radiation_paper/inputs/beam_dynrad_cks.athinput
dyngr_radiation_paper/inputs/beam_dynrad_adm_flat.athinput

```

The maximum absolute difference in R^{00} between the legacy CKS beam and the standalone CKS beam is zero in the current regression. The maximum absolute difference between the legacy CKS beam and the ADM flat-space beam is also zero. In addition to the one-dimensional slice in Figure 1, the same runs now write a two-dimensional x - y binary slice at $z = 0$. The two-dimensional fields are shown in Figure 2, and residual maps against the legacy solver are shown in Figure 3. The maximum

absolute residuals in the 2D maps are 0 for both the CKS-compatible and ADM-flat branches. The same ADM flat beam input also completed under `mpirun -n 2`, testing MPI decomposition for this path.

A separate stationary Kerr-Schild ADM smoke test is included at `inputs/tests/dynrad_bh_beam_adm.athinput`. That input keeps the metric time-independent but non-flat, and runs `geometry=adm` with `angular_fluxes=true`. It therefore exercises the Hamiltonian angular drift, metric-gradient cache, and ADM angular timestep limiter without relying on the BBH problem generator. The short regression completed to $t = 0.2$ with finite radiation output. The excision-aware ADM CFL limiter kept the minimum reported timestep at 4.374670×10^{-2} , rather than letting masked black-hole cells dominate the reduction.

The dynamic-metric smoke input

`inputs/tests/dynrad_z4c_wave_adm.athinput`

adds `dyn_radiation` to the Z4c linear-wave background. This checks that ADM metric fields supplied by the Z4c-to-ADM conversion are accepted by the standalone radiation module and that the per-stage geometry cache can be refreshed on a time-dependent metric with `angular_fluxes=true`. The compact regression completed without NaNs or infinities.

The tracked stress harness `dyngr_radiation_paper/scripts/run_stress_tests.py` runs the compact serial, restart, MPI, and dynbbh cases with outputs under `/tmp`. In the current run it passed all 26 checks: fourteen serial solver inputs, five two-rank MPI inputs, the write/resume restart pair, the expected fatal rejection of `dyn_radiation geometry=cks` with `z4c`, the expected dynbbh rejection of legacy `radiation`, two serial dynbbh inputs, and one two-rank dynbbh smoke test. The harness scans text outputs and logs for non-finite values. Additional fixed-fluid checks verified that `fixed_fluid=true`, `affect_fluid=false` runs successfully and that `fixed_fluid=true`, `affect_fluid=true` fails at input validation. The added serial solver-quality inputs check conservative angular positivity redistribution in both CKS and ADM modes and the iterated temperature-dependent source solve. The current matrix also includes the ADM-specific FLRW redshift, static lapse-gradient, and momentum-source closure tests described below.

7.2 Crossing beams

The crossing-beams regression initializes two Gaussian beams entering from the left side of a flat domain and crossing near its center. Because each angular bin is transported independently, the beams should free-stream through one another. The built-in problem generator `rad_crossing_beams` supports both the legacy `radiation` solver and `dyn_radiation`; the committed inputs are `inputs/tests/rad_crossing_beams.athinput` and `inputs/tests/dynrad_crossing_beams.athinput`, plus the analytic ADM variant `inputs/tests/dynrad_crossing_beams_adm.athinput`. The ADM input uses `<adm dynamic=false>` with the Minkowski metric and does not enable Z4c. The helper script `dyngr_radiation_paper/scripts/run_crossing_beams.py` runs the legacy CKS, `dyn_radiation` CKS, and `dyn_radiation` ADM branches for `nlevel = 1, ..., 6`, plots R^{tt} , and compares against the analytic straight-line Gaussian-beam solution.

For a given angular mesh, the initialized one-sided Gaussian beam profiles start at the marked source circles, follow the requested physical beam axes, and are kept consistent in ghost zones by the user boundary fill. The angular intensities are distributed over all angular bins by a positive maximum-entropy projection. The projection has exact unit zeroth moment and exact first moment along the requested beam axis at the largest realizable positive flux factor of the finite angular grid, multiplied by the default safety factor 0.995. This avoids nearest-angle launch bias while keeping all primitive intensities nonnegative. The numerical and exact maps therefore converge

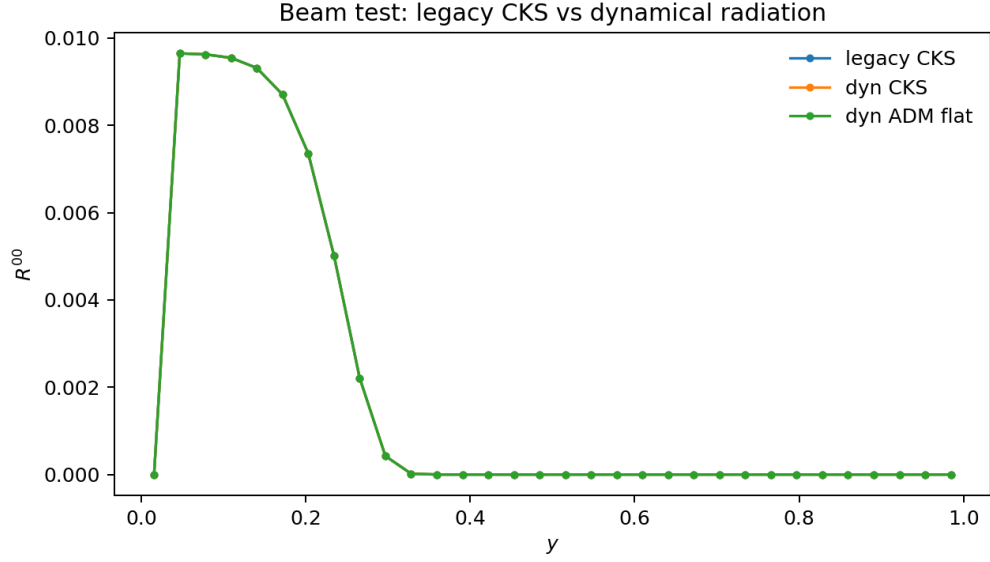


Figure 1: Radiation moment R^{00} along the output slice for the legacy CKS solver, the standalone CKS-compatible branch, and the ADM flat-space branch. The plotted profiles overlap in the current regression.

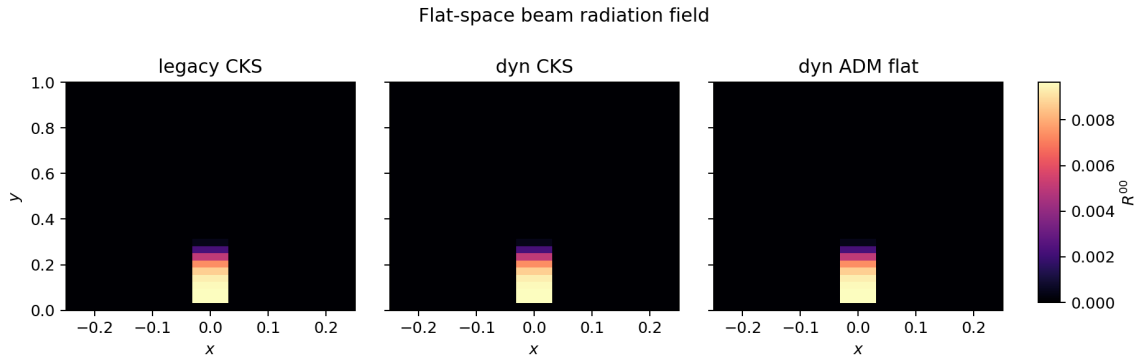


Figure 2: Two-dimensional R^{00} fields for the flat-space beam test. The legacy CKS solver, the standalone CKS-compatible branch, and the ADM flat-space branch use the same color scale.

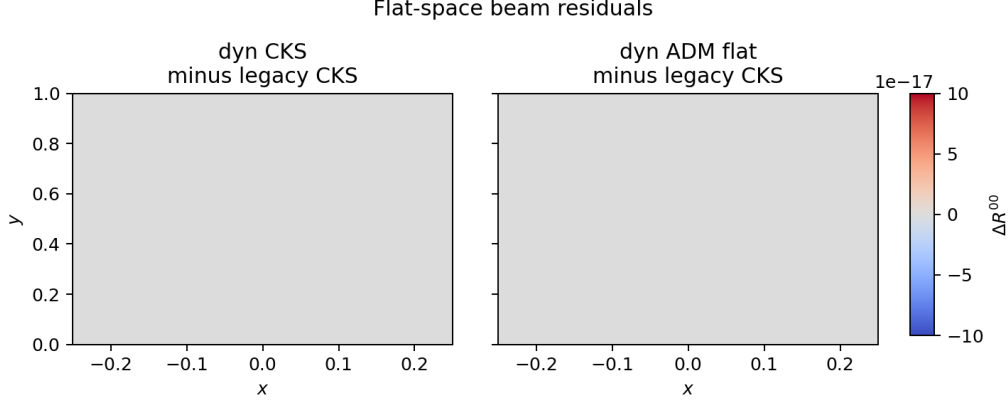


Figure 3: Two-dimensional residuals in R^{00} for the flat-space beam test, computed against the legacy CKS solution. The residuals are zero to the reported precision in the current regression.

as angular resolution improves. In the current run the legacy solver and the CKS-compatible `dyn_radiation` solver agree to the reported precision for every angular resolution. The analytic ADM branch overlays the CKS branch visually; its normalized L_∞ difference from the CKS branch is 3.757567×10^{-4} at $N_{\text{ang}} = 12$ and 2.443248×10^{-5} at $N_{\text{ang}} = 362$. The relative L_1 error against the analytic R^{tt} field decreases from 1.956930×10^{-2} at $N_{\text{ang}} = 12$ to 1.504625×10^{-2} at $N_{\text{ang}} = 362$ for CKS, and from 1.956859×10^{-2} to 1.504618×10^{-2} for ADM. The remaining error is set by the finite angular direction set and spatial advection diffusion.

7.3 Kerr photon-orbit beam

The black-hole beam regression uses the same stationary Kerr-Schild geometry in the legacy solver, the CKS-compatible `dyn_radiation` branch, and the analytic ADM `dyn_radiation` branch. The ADM input uses `<adm dynamic=false>` with the Kerr-Schild metric and does not enable Z4c. The built-in problem generator `rad_kerr_orbit_beam` enrolls a user radiation source term rather than the legacy cone source. At setup it computes the equatorial counter-rotating Kerr photon-orbit radius for the requested spin, maps the tangent null direction into the local tetrad at the source, and projects that direction onto a positive angular stencil. The source then adds intensity with unit angular zeroth moment and first moment parallel to the photon-orbit tangent. This avoids a nearest-angle launch error while retaining a compact local setup with $n_{x_3} = 1$.

The committed inputs are `inputs/tests/rad_kerr_orbit_beam.athinput` and `inputs/tests/dynrad_kerr_orbit_beam.athinput`, plus the ADM variant `inputs/tests/dynrad_kerr_orbit_beam_adm.athinput`. The helper script `dyngr_radiation_paper/scripts/run_kerr_orbit_beam.py` runs all three branches and overlays the analytic equatorial photon-orbit guide. With `nlevel = 4` ($N_{\text{ang}} = 162$) and a $72 \times 72 \times 1$ mesh, the normalized infinity-norm difference between the CKS `dyn_radiation` map and the legacy CKS map is 7.298162×10^{-2} after $t = 5$, with a relative L_1 difference of 1.595626×10^{-2} . The ADM map follows the same photon-orbit guide; its normalized infinity-norm difference from the CKS `dyn_radiation` map is 4.497598×10^{-2} , with a relative L_1 difference of 7.477185×10^{-2} . The largest discrepancies occur along weak, angularly diffused beam shoulders; the high-intensity ridge follows the same guide in all three branches.

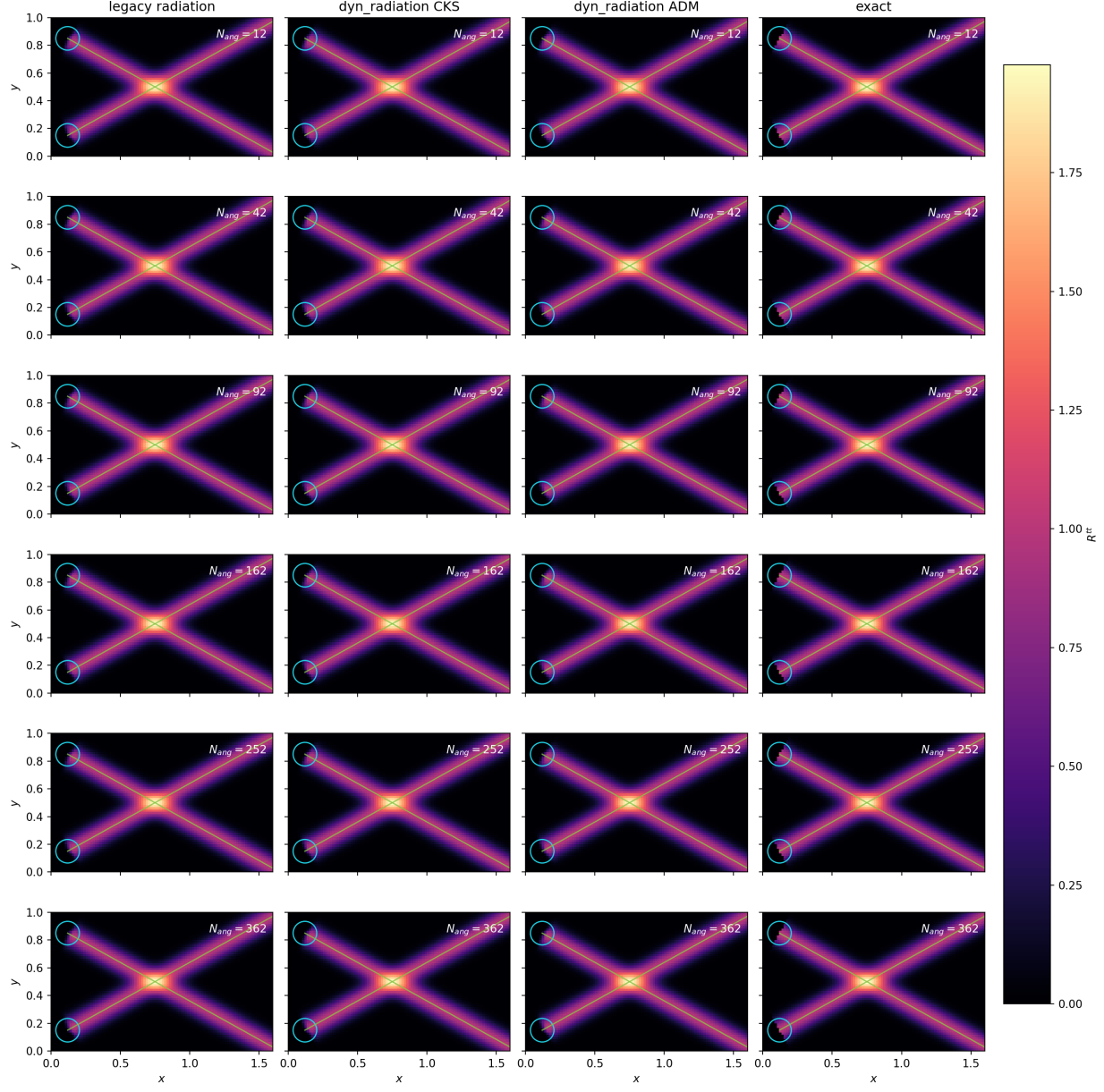


Figure 4: Crossing-beams R^{tt} maps for the legacy solver, `dyn_radiation` in CKS compatibility mode, `dyn_radiation` with analytic Minkowski ADM data, and the analytic solution. The numerical columns overlap in the current regression and approach the analytic beam geometry as N_{ang} increases.

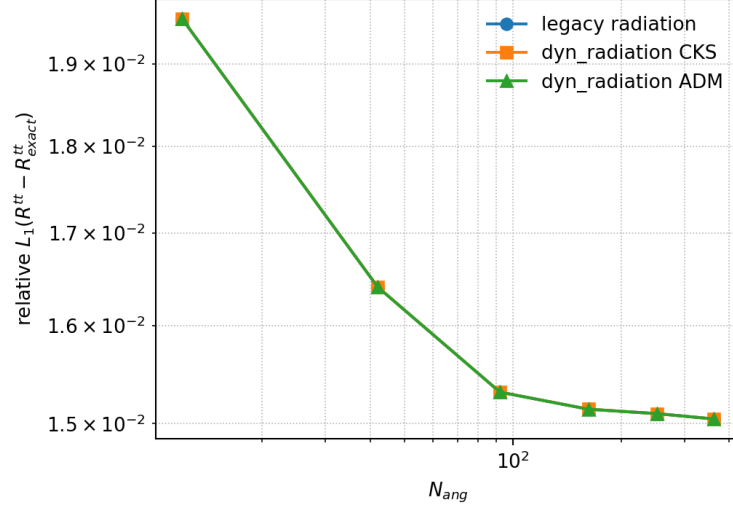


Figure 5: Relative L_1 error of the crossing-beams R^{tt} field against the analytic straight-line Gaussian-beam solution as the angular mesh is refined.

7.4 Gas-radiation equilibration

The homogeneous equilibration regression initializes gas and radiation out of thermal equilibrium and evolves only the local absorption/emission source. The built-in problem generator `rad.equilibration` supports both `radiation` and `dyn_radiation`; the paired inputs are `inputs/tests/rad_equilibration.athinput` and `inputs/tests/dynrad_equilibration.athinput`. The default setup uses $\rho = 1$, $\Gamma = 5/3$, $T_{\text{gas}}(0) = 2$, $T_{\text{rad}}(0) = 1$, $a_{\text{rad}} = 1$, pure absorption $\kappa_a = 1$, and a homogeneous periodic mesh. With $c_V = \rho/(\Gamma - 1)$, the conserved total energy density is

$$u_{\text{tot}} = c_V T_{\text{gas}} + a_{\text{rad}} T_{\text{rad}}^4 = 4.$$

The exact comparison curve solves the scalar ODE

$$c_V \frac{dT_{\text{gas}}}{d\tau} = - (a_{\text{rad}} T_{\text{gas}}^4 - [u_{\text{tot}} - c_V T_{\text{gas}}]), \quad T_{\text{rad}} = \left(\frac{u_{\text{tot}} - c_V T_{\text{gas}}}{a_{\text{rad}}} \right)^{1/4},$$

where $\tau = \kappa_a t$. The script `dyngr_radiation_paper/scripts/run_equilibration.py` runs both solvers with effective 1-step, 10-step, and 100-step source integrations to $\tau = 1$, overlays the exact ODE solution, and checks conservation of u_{tot} . In the current run the 100-step `dyn_radiation` solution had maximum errors $\max |\Delta T| = 1.712653 \times 10^{-2}$ and $\max |\Delta u| = 2.568980 \times 10^{-2}$ against the exact curve. The final states of the legacy and `dyn_radiation` runs agreed: $T_{\text{gas}} = 1.216323$, $T_{\text{rad}} = 1.214481$, and $u_{\text{tot}} = 4.000000$.

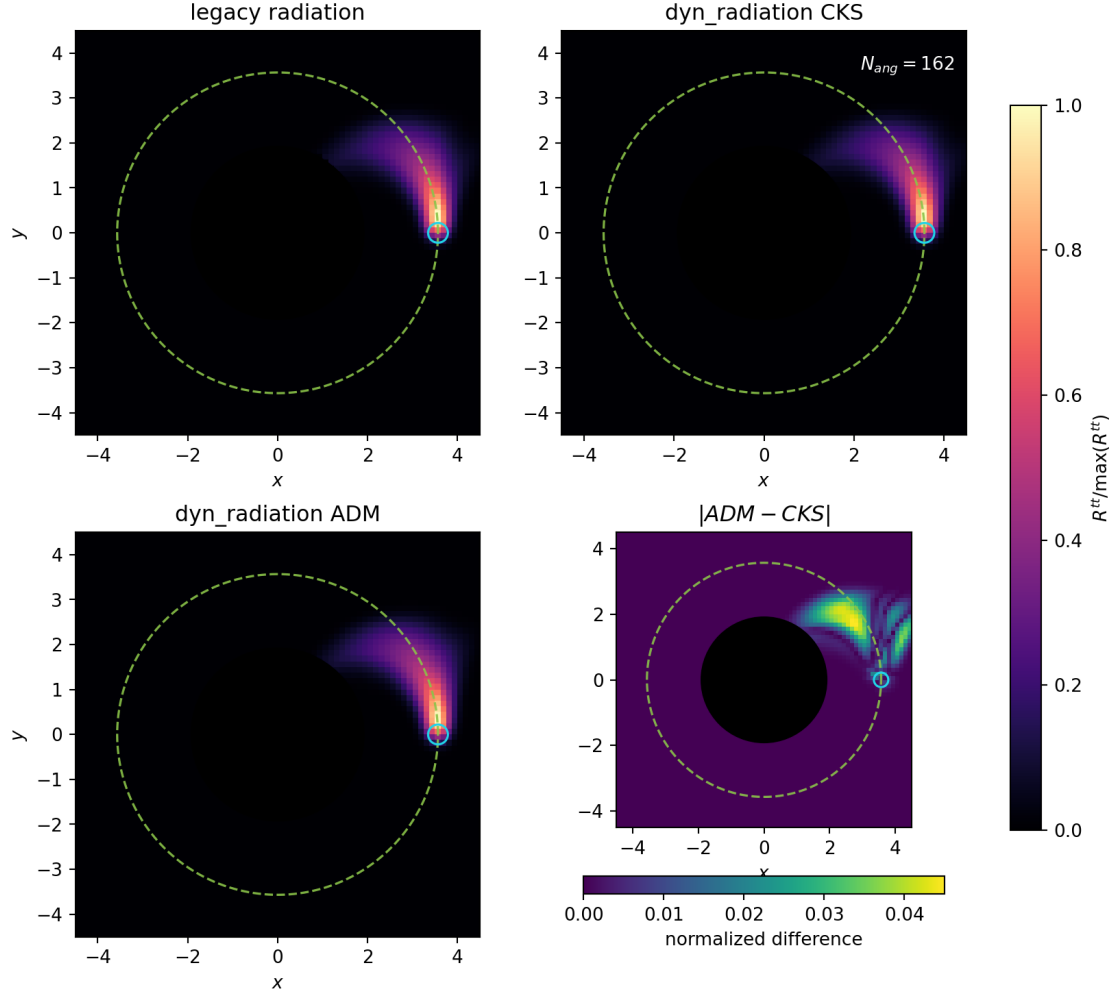


Figure 6: Kerr photon-orbit beam comparison. The dashed green curve is the equatorial photon-orbit guide, the cyan circle marks the projected source region, and the black disk marks the horizon-scale excision region. The panels compare the legacy CKS solver, `dyn_radiation` CKS, analytic ADM `dyn_radiation`, and the normalized ADM-CKS difference on the same compact $n_{x_3} = 1$ grid.

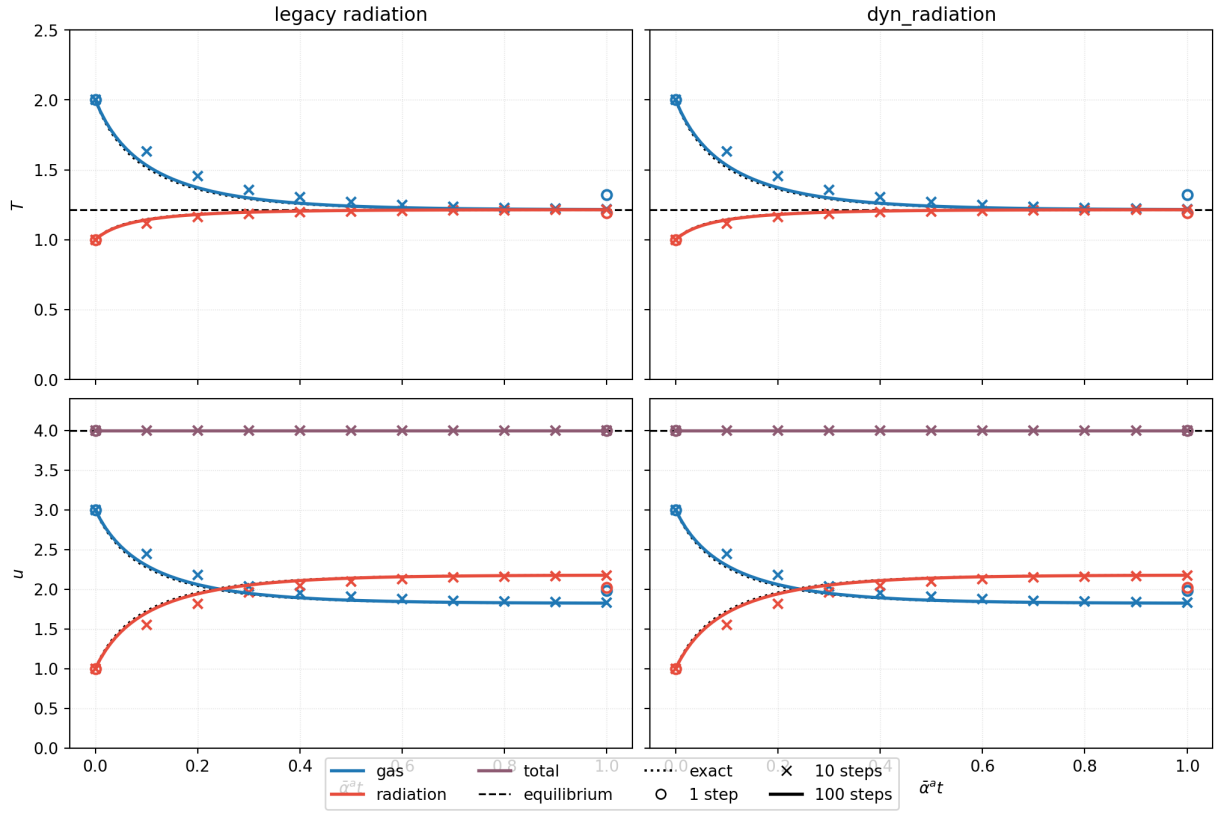


Figure 7: Homogeneous gas-radiation equilibration for the legacy solver and `dyn_radiation`. Symbols show coarse source-step integrations, solid curves show the 100-step run, dashed lines mark the equilibrium state, and dotted curves show the exact energy-conserving relaxation ODE.

7.5 Radiation-hydrodynamic linear wave

The stationary CKS compatibility branch was also checked against the legacy radiation solver on a radiation-hydrodynamic linear wave. The final error rows match:

$$\begin{aligned}
 \text{RMS-}L_1 &= 2.400907 \times 10^{-6}, \\
 d_{L_1} &= 1.685676 \times 10^{-6}, \\
 M_{L_1}^1 &= 1.380892 \times 10^{-6}, \\
 E_{L_1} &= 1.007962 \times 10^{-6}.
 \end{aligned} \tag{75}$$

The corresponding inputs are `dyngr_radiation_paper/inputs/lwave_radiation_cks.athinput` and `dyngr_radiation_paper/inputs/lwave_dynrad_cks.athinput`.

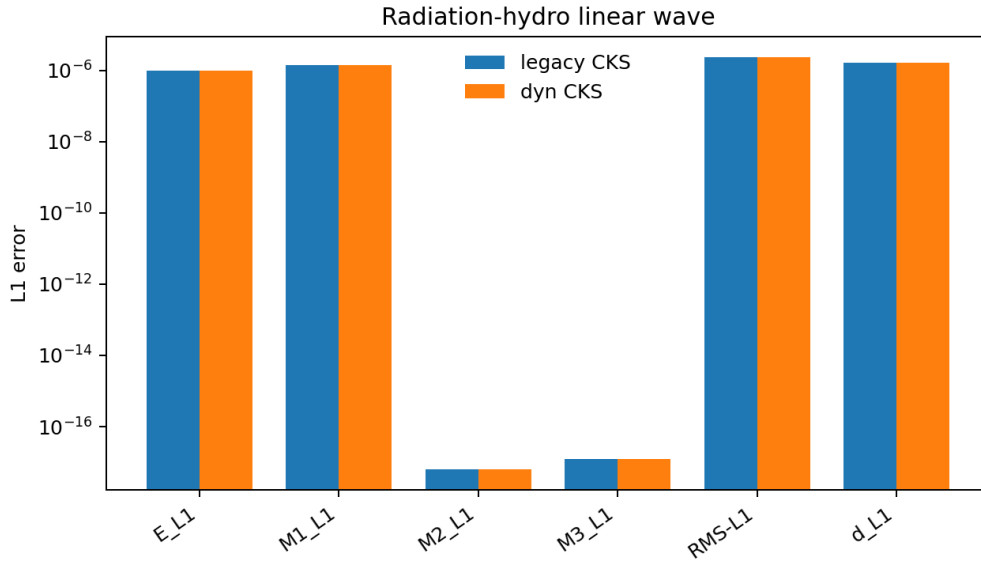


Figure 8: Linear-wave error metrics for the legacy solver and `dyn_radiation` in stationary CKS compatibility mode.

The convergence form of the same test is generated by `dyngr_radiation_paper/scripts/run_linear_wave_convergence.py` from `inputs/tests/rad_lwave_convergence.athinput`, `inputs/tests/dynrad_lwave_convergence.athinput`, and `inputs/tests/dynrad_lwave_adm_convergence.athinput`. The ADM input uses `<adm dynamic=false>` with the analytic Minkowski ADM metric and a zero-field `<mhd>` wrapper, because ADM/Z4c backgrounds are connected to the fluid package through `dynGRMHD` rather than `<hydro>`. These runs use $\delta = 10^{-4}$, which is small enough to remain in the linear regime but large enough that the final binary outputs are not limited by single-precision output quantization. The script measures normalized L_1 errors in ρ , u^x , P_{gas} , R^{tt} , and R^{tx} directly against the analytic eigenmode at $N_{\text{cell}} = 16, 32, 64, 128, 256$. The legacy, CKS `dyn_radiation`, and analytic-flat ADM `dyn_radiation` curves overlap to plotting precision. The last-three-point convergence rates in the current run were

	RMS	ρ	u^x	P_{gas}	R^{tt}	R^{tx}
dyn CKS	2.071	1.849	2.129	2.111	2.121	2.247
dyn ADM	2.070	1.849	2.129	2.105	2.121	2.247

which is consistent with the second-order reconstruction and RK2 time integration used in the input. The density rate is slightly lower over this finite range because that component is closest to the binary-output precision floor at the highest resolution.

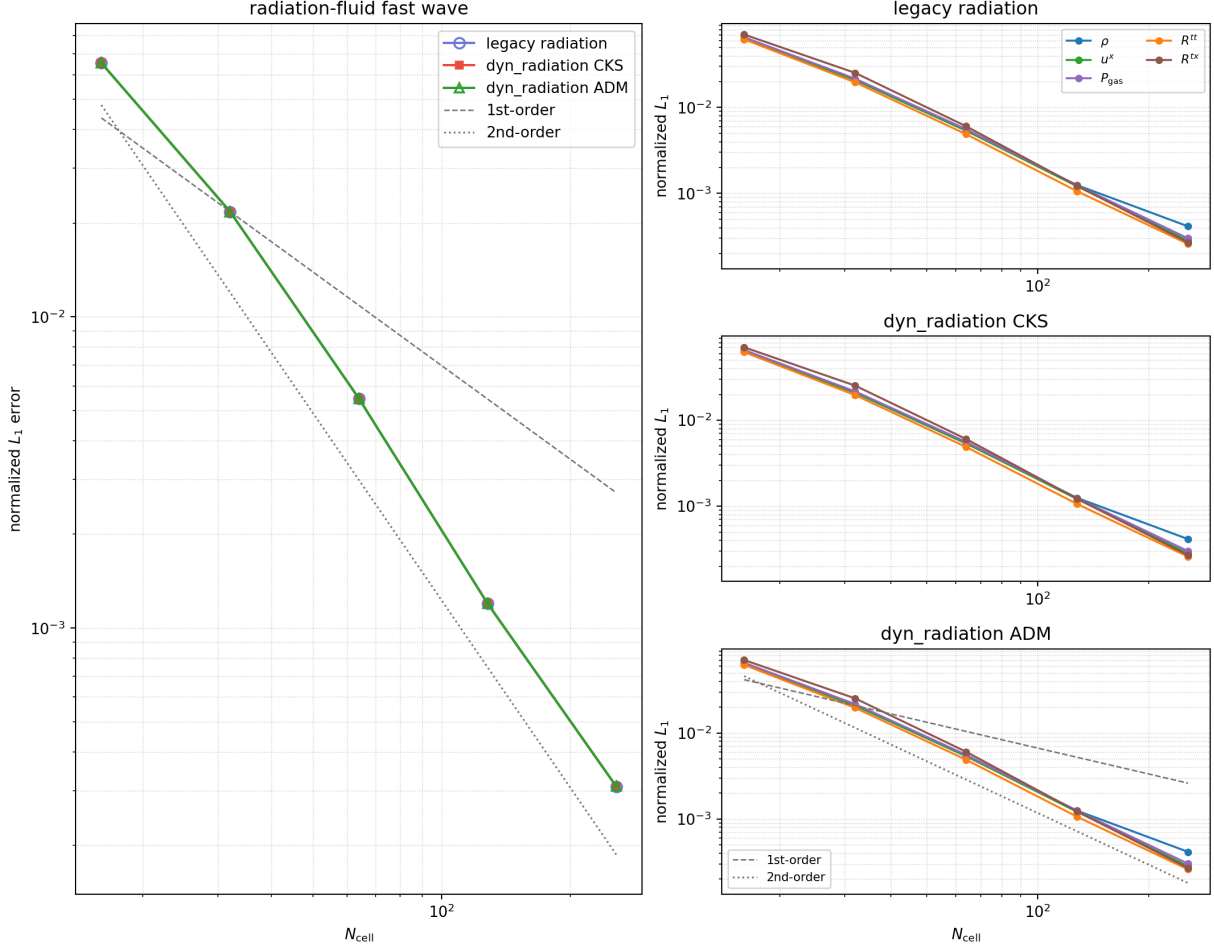


Figure 9: Radiation-fluid linear-wave convergence. The left panel shows the normalized coupled L_1 error for the legacy CKS solver, `dyn_radiation` in CKS mode, and `dyn_radiation` in analytic-flat ADM mode; the curves overlap. The right panels split the error by fluid and radiation-moment components for each solver path.

7.6 ADM formal regressions

Three additional compact regressions target the ADM-only pieces that are not isolated by the flat CKS compatibility tests. The committed inputs are `inputs/tests/dynrad_flrw_redshift.athinput`, `inputs/tests/dynrad_lapse_gradient.athinput`, and `inputs/tests/dynrad_momentum_source.athinput`; the helper script `dyngr_radiation_paper/scripts/run_adm_formal_tests.py` reruns them and writes Figure 10. These tests use `<adm dynamic=false>` with analytic ADM data supplied by the problem generator, so they exercise the ADM radiation geometry path without invoking Z4c.

The FLRW redshift test sets $\alpha = 1$, $\beta^i = 0$, $\gamma_{ij} = a(t)^2 \delta_{ij}$, and $K_{ij} = -a \dot{a} \delta_{ij}$, with $a(t) = 1 + 0.2t$. A homogeneous isotropic radiation field should satisfy $E \propto a^{-4}$ and $\sqrt{\gamma}E \propto a^{-1}$. The current run to $t = 0.5$ gave $E = 0.6829324379755602$ against $a^{-4} = 0.6830134553650705$, and $\sqrt{\gamma}E =$

0.908983074945471 against $a^{-1} = 0.9090909090909091$; both relative errors were 1.18618×10^{-4} . This checks the sign of $\alpha K_{ij} s^i s^j$ in the geometric energy source.

The static lapse-gradient test sets $\alpha(x) = 1 + 0.1 \sin(2\pi x)$, $\beta^i = 0$, and $\gamma_{ij} = \delta_{ij}$, and initializes a uniform anisotropic radiation field with positive x -directed flux. For the short run $t = 0.01$, the first-order source prediction is $\Delta E = -2E_0 f_x(\partial_x \alpha)t$. The numerical perturbation has correlation 0.999307 with that analytic shape and relative RMS mismatch 4.68375×10^{-2} , consistent with using the full transport update rather than an isolated infinitesimal source step.

The momentum-source closure test compares the angular sum of the ADM Hamiltonian force, $\sum_n w_n I_n A_j$, to the Valencia momentum source

$$-E \partial_j \alpha + S_i \partial_j \beta^i + \frac{1}{2} \alpha P^{ik} \partial_j \gamma_{ik}.$$

The two sides are evaluated from the same cached ADM gradients used by the solver. The maximum absolute mismatch decreased from 1.80196×10^{-5} at $N_x = 32$ to 1.14669×10^{-6} at $N_x = 128$, following the expected second-order finite-difference cache error. The maximum relative mismatch at $N_x = 128$ was 8.47848×10^{-4} ; that relative norm is dominated by cells where the source denominator is small.

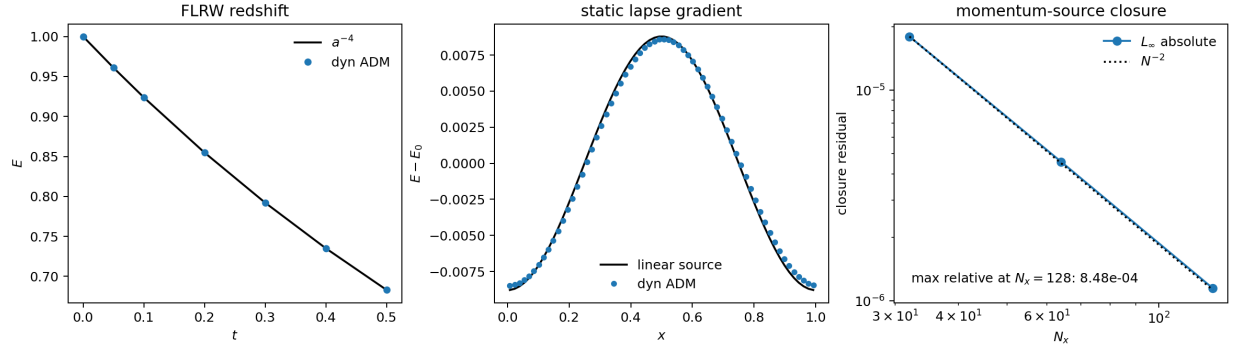


Figure 10: ADM-specific formal regressions. Left: homogeneous FLRW radiation redshifts as $E \propto a^{-4}$. Center: the static lapse-gradient source has the predicted sign and shape. Right: the Hamiltonian angular-force sum and the Valencia momentum source agree with a second-order absolute residual as the ADM metric-gradient cache is refined.

7.7 Binary-background beam and particle smoke test

The binary-background input is

```
dyngnr_radiation_paper/inputs/dynbbh_beam_particles.athinput.
```

It exercises the ADM beam source on the superposed boosted Kerr-Schild binary background in `src/pgen/dynbbh.cpp`. It runs with `geometry=adm`, `angular_fluxes=true`, and `adm/dynamic=true`; the latter is required because the analytic superposed-BBH ADM callback depends on `pmesh->time`. The dynbbh initializer now uses only `pdynrad`; the legacy `prad` path is rejected because its stationary CKS normalization is inconsistent with the ADM background. It also initializes null particles on the beam edge and advances them using Equations (73) and (74). The committed input remains a short smoke test ($t = 0.2$); the figure runner `dyngnr_radiation_paper/scripts/run_dynbbh_beam_figures.py` overrides the same input for plotting. The current high-resolution figure run uses a $72 \times 72 \times 4$ mesh, $36 \times 36 \times 4$ MeshBlocks, N_{ang} from `nlevel=4`, four MPI ranks, a 20° source

cone, and advances to $t = 10$. This matches the single-BH angular resolution while keeping the minimum active x^3 resolution accepted by the mesh driver. It completed in 181 cycles with empty stderr; the final lineout has $\max R^{tt} = 6.247600$ and lineout integral 6.491427. The dynbbh pgen drops non-source-rank placeholder particles in MPI and retags the remaining source particles, so the tracked-particle output follows only real null beam tracers. The radiation source is a fixed coordinate-time, fixed coordinate-direction injection; the dashed guide in the figures marks this source axis rather than an expected geodesic in the curved, time-dependent background. The radiation maps, ADM lapse/conformal-factor background, null beam-edge trajectories, and time-series snapshots are shown in Figures 11, 12, and 13. A separate serial run enabled ADM radiation-matter coupling with absorption and scattering opacities and completed without NaNs. These are compact runtime tests; long production dynbbh/MHD radiation evolutions remain a separate validation target.

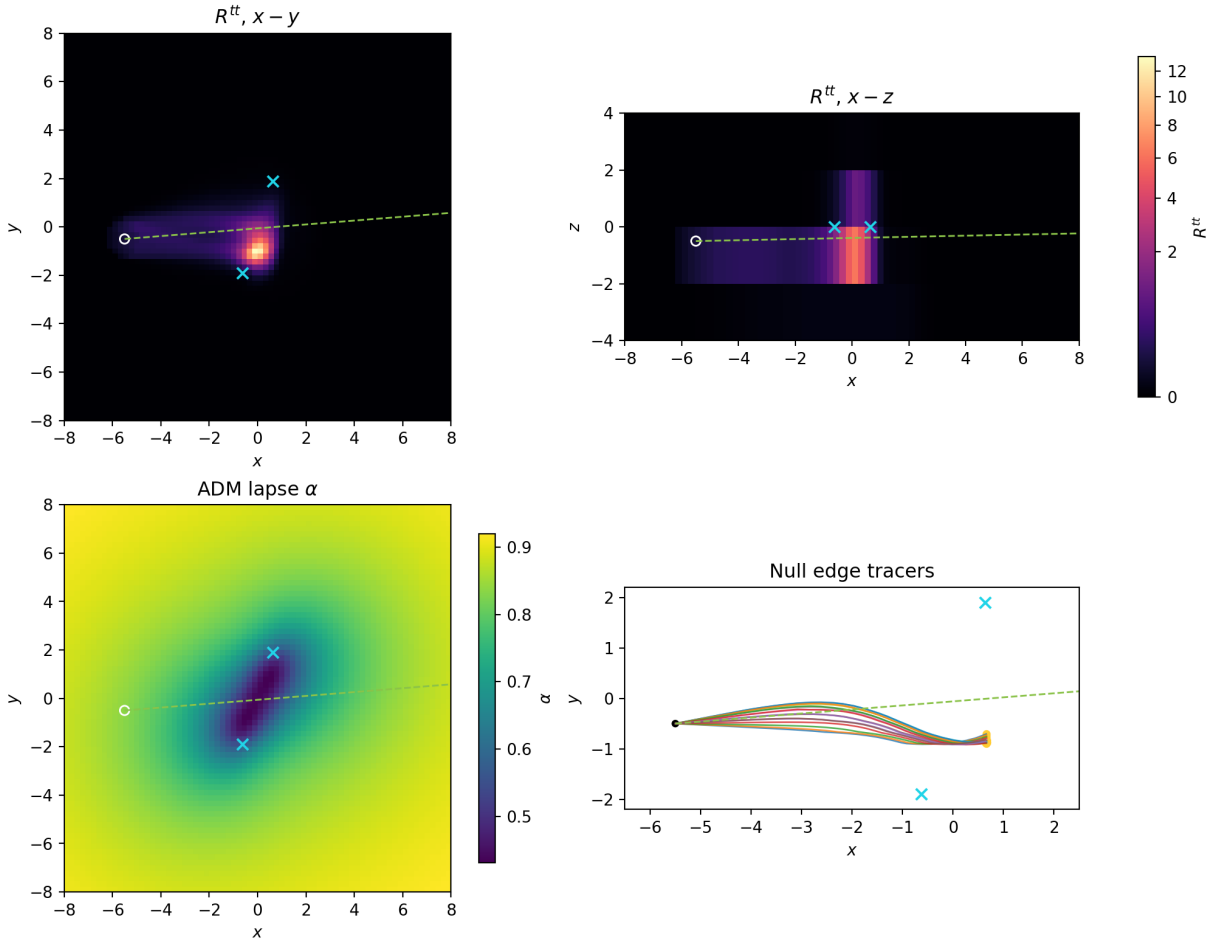


Figure 11: Compact dynbbh ADM beam diagnostic on the superposed orbiting Kerr-Schild background. The panels show R^{tt} in $x - y$ and $x - z$ slices, the ADM lapse on the same binary slice, and null particle trajectories initialized on the edge of the beam cone.

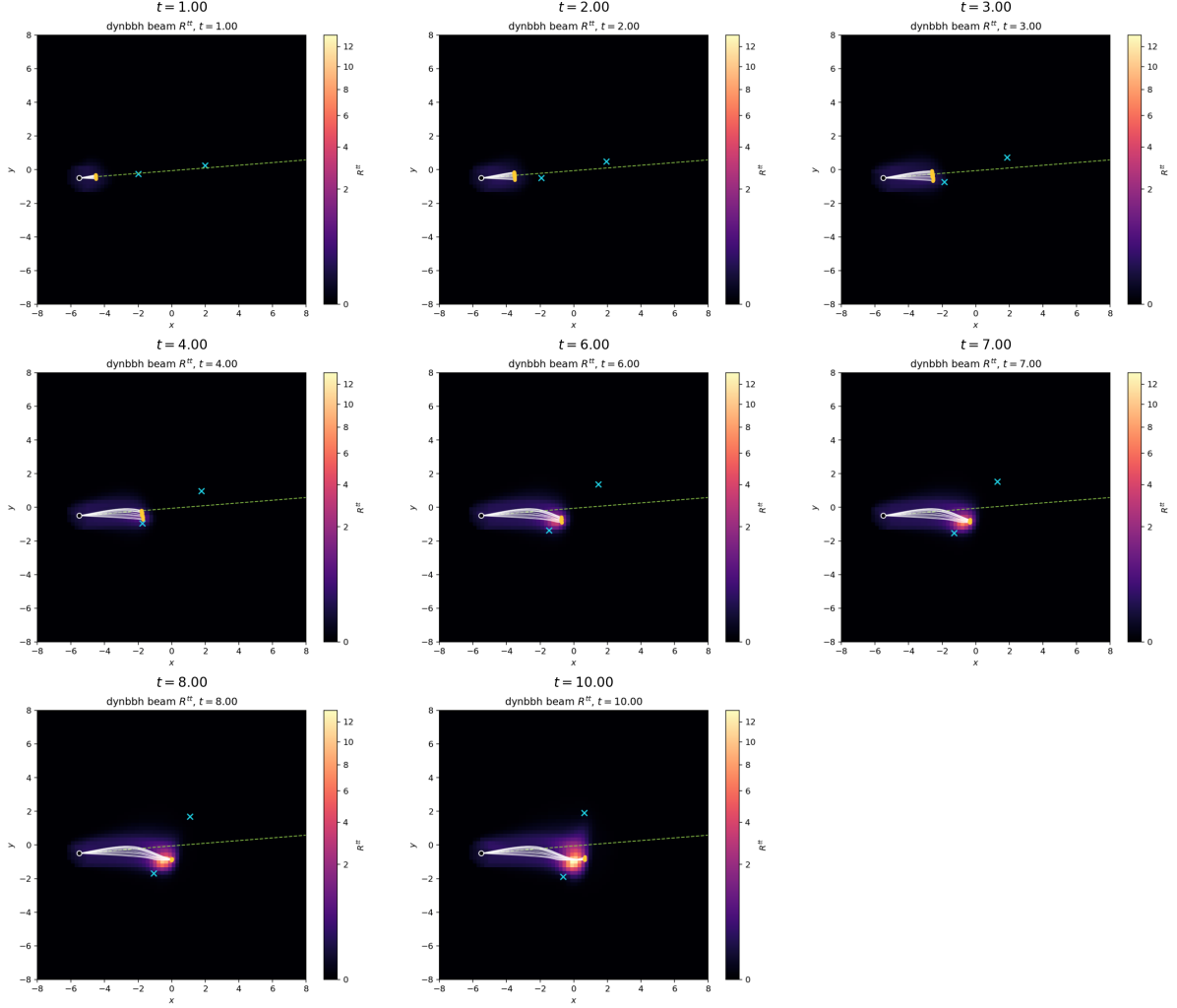


Figure 12: High-resolution dynbbh ADM beam time series. Each panel shows the $x - y$ slice of R^{tt} with the accumulated null-particle trajectories overplotted through that output time. The binary centers rotate according to the superposed orbiting Kerr-Schild background used to populate the analytic ADM fields.

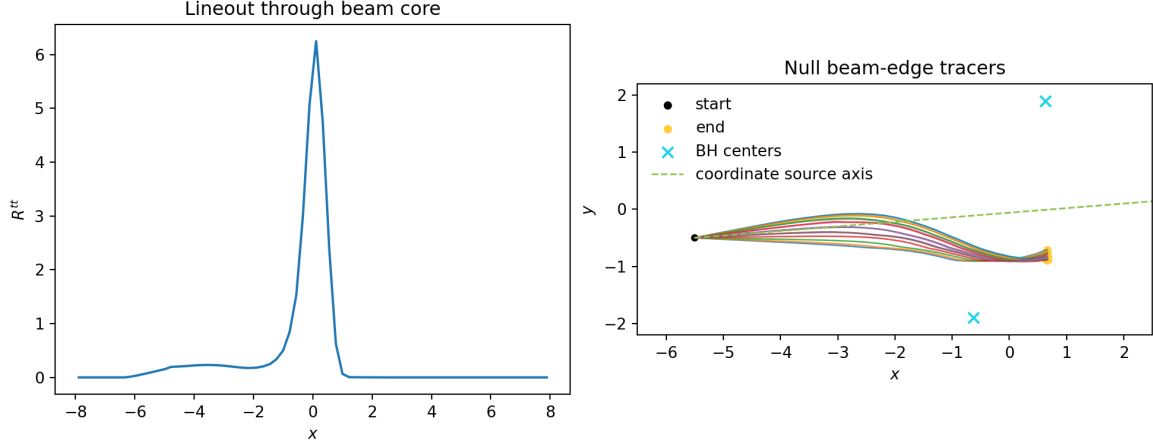


Figure 13: Beam-core R^{tt} lineout and null beam-edge particle tracks for the same compact dynbbh figure run.

7.8 Diagnostic artifacts

All diagnostic figures in this section are generated from committed inputs and scripts. Table 2 lists the plot artifacts, their producers, and the numerical behavior each one is intended to check.

8 Comparison With the Existing Solver

The key mathematical difference is the removal of the signed stationary-metric normalization in ADM mode. The key implementation difference is that metric data are no longer assumed to be analytic or time independent. The new solver therefore fits the same design pattern as the Valencia GRMHD path: fluxes and sources are built from ADM variables supplied by the active spacetime module. At present this analogy is limited to transport and matter coupling. The radiation stress-energy tensor is not passed to the metric solver, so Z4c evolutions see MHD/dynGRMHD matter sources only.

9 How to Check the Formalism

The derivation above gives several concrete checks that can be applied to the implementation.

Flat-space limit. Set $\alpha = 1$, $\beta^i = 0$, $\gamma_{ij} = \delta_{ij}$, and $K_{ij} = 0$. Then $E^i_{(\hat{a})} = \delta^i_{\hat{a}}$, $U_n = I_n$, $v^i = s^i$, $S_n^{\text{geom}} = 0$, and all angular drift coefficients vanish. The ADM solver must reduce to ordinary Cartesian free-streaming. This is the reason the flat ADM beam can match the legacy flat CKS beam exactly.

Moment-source check. For any angular distribution, compute the discrete moments in Equation (25). The angular sum of the per-angle energy source in Equation (29) must equal the right-hand side of Equation (27). The automated `rad_momentum_source` regression now performs the stronger local check that the angular Hamiltonian force sum produces the Valencia momentum source in Equation (28). The residual is expected to be second order in the ADM metric-gradient cache spacing because $\partial_j \gamma_{ik}$ and $\partial_j \gamma^{ik}$ are independently finite-differenced.

Figure file	Producer	Diagnostic
figures/beam_comparison.png	plot_comparisons.py	1D flat beam agreement across legacy CKS, dyn CKS, and dyn ADM-flat
figures/beam_2d_fields.png	plot_comparisons.py	2D flat beam morphology in the same three branches
figures/beam_2d_residuals.png	plot_comparisons.py	residual maps against the legacy flat beam
figures/linear_wave_errors.png	plot_comparisons.py	final scalar error row for the CKS linear wave
figures/linear_wave_convergence.png	run_linear_wave_convergence.py	coupled fluid/radiation linear-wave convergence for legacy CKS, dyn CKS, and analytic-flat dyn ADM
figures/crossing_beams_comparison.png	run_crossing_beams.py	free streaming through crossing beams for legacy CKS, dyn CKS, dyn ADM-flat, and the analytic solution
figures/crossing_beams_convergence.png	run_crossing_beams.py	angular convergence for crossing beams in all three branches
figures/kerr_orbit_beam_comparison.png	run_kerr_orbit_beam.py	compact Kerr photon-orbit beam bending, including analytic ADM Kerr-Schild transport
figures/equilibration_comparison.png	run_equilibration.py	homogeneous gas-radiation relaxation against the exact ODE
figures/adm_formal_tests.png	run_adm_formal_tests.py	FLRW redshift, static lapse-gradient source, and ADM momentum-source closure
figures/dynbbh_beam_particles.png	run_dynbbh_beam_figures.py	ADM BBH beam source and null-particle beam-edge diagnostic
figures/dynbbh_beam_rtt_slices.png	run_dynbbh_beam_figures.py	R^{tt} beam slices on the superposed orbiting Kerr-Schild background
figures/dynbbh_beam_adm_background.png	run_dynbbh_beam_figures.py	ADM lapse and conformal-factor contours for the same binary slice
figures/dynbbh_beam_particles_lineout.png	run_dynbbh_beam_figures.py	beam-core lineout and null beam-edge tracers
figures/dynbbh_beam_summary.png	run_dynbbh_beam_figures.py	combined dynbbh beam/background/tracer diagnostic
figures/dynbbh_beam_timeseries.png	run_dynbbh_beam_figures.py	high-resolution dynbbh time-series snapshots with accumulated null-particle tracks

Table 2: Diagnostic plot artifacts for the standalone dynamical radiation validation suite.

Test	Quantity checked	Result
Flat beam, CKS compatibility	$\max R_{\text{dynrad}}^{00} - R_{\text{legacy}}^{00} $	0
Flat beam, ADM mode	$\max R_{\text{ADM}}^{00} - R_{\text{legacy}}^{00} $	0
Flat ADM beam, MPI	two-rank decomposition	passed
Stationary KS ADM beam	nonzero ADM angular drift	passed
Dynamic Z4c ADM smoke	time-dependent ADM metric cache	passed
CKS linear wave	final error row	identical
CKS/ADM linear-wave convergence	fluid and radiation moment errors versus analytic eigenmode	legacy/dyn CKS/dyn ADM overlap; RMS rate 2.07
Crossing beams	free-streaming through another beam and angular convergence	source-aligned from $N_{\text{ang}} = 12$; ADM-CKS $L_{\infty} \leq 4.3 \times 10^{-4}$
Kerr photon-orbit beam	stationary Kerr bending on an $n_{x_3} = 1$ grid	legacy, CKS, and ADM ridges follow orbit; ADM-CKS $L_{\infty} = 4.50 \times 10^{-2}$
Gas-radiation equilibration	homogeneous source relaxation and total-energy conservation	final legacy/dyn states identical; $u_{\text{tot}} = 4$
FLRW ADM redshift	homogeneous $E \propto a^{-4}$, $\sqrt{\gamma}E \propto a^{-1}$	final relative errors 1.19×10^{-4}
Static lapse-gradient source	sign and shape of $-F^i \partial_i \alpha$	correlation 0.999307, relative RMS 4.68×10^{-2}
ADM momentum-source closure	Hamiltonian force sum versus Valencia momentum source	absolute residual 1.15×10^{-6} at $N_x = 128$
Positivity limiter	angular zeroth-moment preserving redistribution (CKS and ADM)	passed
Iterated source solve	temperature-dependent fixed-fluid coupling	passed
Restart	write and resume <code>dynrad_beam_cks</code>	passed
Z4c CKS rejection	unsupported metric mode guard	passed
BBH legacy radiation rejection	<code>prad</code> guard	passed
BBH ADM beam	finite beam and real source-rank particle tracks	passed in four-rank MPI to $t = 10$
BBH ADM coupling	absorption/scattering with MHD primitives	passed in serial
BBH coupled MPI smoke	two-rank dynbbh/MHD run	passed

Table 3: Current regression status for the standalone dynamical radiation implementation.

Feature	radiation	dyn_radiation
Metric source	Analytic stationary CKS metric	CKS compatibility mode or ADM mesh fields
Conserved intensity	signed $k^0 k_0 I$	CKS: signed $k^0 k_0 I$; ADM: $\sqrt{\gamma} I$
Spatial speed	k^i/k^0 from stationary tetrad	CKS: k^i/k^0 ; ADM: $\alpha s^i - \beta^i$
Metric source terms	Stationary tetrad and angular drift coefficients	ADM Valencia energy source using K_{ij} and $\partial_i \alpha$
Angular bending	Precomputed CKS coefficients	Cached ADM Hamiltonian angular drift
Matter coupling	Existing local implicit path	CKS path retained; ADM uses $I = U/\sqrt{\gamma}$, Eulerian tetrad frequency, and Valencia moment feedback
Metric backreaction	Not applicable to fixed metric	Not included; radiation is metric-passive
Binary backgrounds	Not supported directly	Supported through ADM fields

Table 4: Functional comparison between the existing stationary radiation solver and the new standalone dynamical-background solver.

ADM redshift checks. The `rad_flrw_redshift` regression verifies the sign of the $\alpha K_{ij} s^i s^j$ source by checking $E \propto a^{-4}$ and $\sqrt{\gamma} E \propto a^{-1}$. The `rad_lapse_gradient` regression verifies the sign of $-s^i \partial_i \alpha$ by comparing a short anisotropic run against the first-order lapse-gradient source prediction.

Tetrad consistency check. At every cached cell,

$$\gamma_{ij} E^i_{(\hat{a})} E^j_{(\hat{b})} = \delta_{\hat{a}\hat{b}}, \quad e^{(\hat{a})}_i E^i_{(\hat{b})} = \delta^{\hat{a}}_{\hat{b}}. \quad (76)$$

The same Cholesky gauge must be used for spatial transport, angular drift, and matter coupling; otherwise spurious angular drift appears even in locally flat coordinates.

Frequency-transform check. For a fluid comoving with the Eulerian observer, $u^{(\hat{0})} = 1$ and $u^{(\hat{a})} = 0$, so $D = \nu_{\text{cm}}/\nu_{\text{E}} = 1$. The ADM coupling must then use $I_{\text{cm}} = U/\sqrt{\gamma}$. For a moving fluid, the fourth-power intensity transform and the inverse-square solid-angle transform in Equation (63) must be applied together; using one without the other gives the wrong comoving mean intensity.

Local conservation check. When `affect_fluid=true`, the matter-coupling kernel should change gas energy and momentum by the negative of the change in the radiation moments in Equation (67). This is independent of how the implicit temperature solve is performed.

10 Discussion and Limitations

The present implementation is a controlled step toward dynamical radiation transport. It moves radiation energy on ADM backgrounds, bends the angular directions with an ADM Hamiltonian operator, reproduces stationary beam and linear-wave regressions, and diagnoses beam propagation on the superposed binary metric. It is still not a final production radiation module. The current coupling to dynamical spacetime is one-way: ADM/Z4c geometry drives radiation, but radiation

does not feed $T_{\mu\nu}$ back into the Z4c right-hand side. The angular operator uses second-order metric-gradient caches; higher-order gradients and limiters can be added behind the same cache interface. The local momentum-source consistency check in Equation (28) is now included in the regression suite; broader momentum-balance tests on genuinely time-dependent ADM backgrounds remain future validation work.

The ADM matter-coupling path now uses the Eulerian tetrad frequency factor, $I = U/\sqrt{\gamma}$, and gas feedback through the Valencia radiation moments. It has passed flat and BBH smoke tests, a homogeneous gas-radiation relaxation comparison against the exact source ODE, and a temperature-dependent fixed-fluid source iteration test. It still needs a broader validation suite with moving-fluid absorption/scattering, optically thick diffusion, and time-dependent metric backgrounds. The ADM metric source is now applied with the exponential source increment in Equation (57). The transport and source updates use the zeroth-moment-conserving angular positivity limiter described above; production diagnostics should still count limiter activations so runs can identify under-resolved optically thin features.

The changed kernels were reviewed for Kokkos device compatibility: the stage geometry cache, timestep reductions, fluxes, and coupling kernels capture only device views and POD scalars, and device-side finiteness tests use `Kokkos::isfinite`. The tests reported here were run on the local CPU/MPI builds. No CUDA, HIP, or SYCL runtime validation was available in this pass, so GPU execution should be treated as statically reviewed but not runtime-certified.

Finally, the null-particle pusher is currently a diagnostic tool. It uses nearest-cell ADM fields and centered finite differences. It is useful for beam edge visualization and for catching obvious metric or sign errors. Its timestep cap uses the ADM coordinate light-speed bound $|\beta^q| + \alpha\sqrt{\gamma^{qq}}$, so it respects shifted and small-lapse coordinates during particle boundary exchange. A scientific geodesic integrator should still use higher-order metric interpolation, adaptive timestepping, and a stricter monitor of the null Hamiltonian.

11 Conclusions

We implemented and documented a standalone dynamical-background radiation module for AthenaK. The solver preserves the legacy CKS variables and flat regression behavior in a compatibility mode and adds an ADM mode based on Eulerian densitized intensity, Valencia coordinate speeds, and a per-angle geometric source whose angular sum recovers the Valencia radiation energy equation. The formulation is suitable for time-dependent ADM backgrounds because metric evolution enters through K_{ij} and the ADM co-triad time derivative, rather than through stationarity assumptions. Regression tests show exact agreement with the legacy flat beam, identical stationary CKS linear-wave error norms, successful two-rank flat ADM beam transport, excision-safe stationary black-hole ADM timesteps, homogeneous gas-radiation equilibration against the exact source ODE, ADM FLRW redshift, ADM lapse-gradient and momentum-source closure checks, restart coverage, and serial plus two-rank ADM beam/null-particle smoke tests on the superposed binary black-hole background. The remaining work is to add radiation metric backreaction, extend ADM validation to moving-fluid and fully time-dependent metric problems, add runtime diagnostics for positivity-limiter activation, and upgrade the diagnostic particle pusher if high-accuracy geodesics are required.